Samuel Boucheron Seguin

Computer Science Engineering Student · CY Tech Pau

Pau, France · on exchange in Gothenburg, Sweden until Jan 2027 samuelbs364@gmail.com github.com/samuelElElohim
linkedin.com/in/samuel-boucheron-seguin samuelelelohim.github.io

> **Seeking:** a full-semester engineering internship in any area of IT, ideally cybersecurity, **16 to 22 weeks between mid-March and August 2027**. Open to France, the Nordics or anywhere in Europe.

SUMMARY

Second-year Computer Science Engineering student at CY Tech with experience from team projects: backend and authentication for a deployed full-stack web platform, and technical lead of a five-person Java project. Particularly interested in cybersecurity.

EDUCATION

CY Tech — Engineering Degree (Ingénieur), Computer Science · Pau, France *2025–2028*

2nd year (ING2) · MSc in Computer Science Engineering expected 2028

University of Gothenburg — Erasmus+ Exchange Semester · Computer Science & Engineering, joint with Chalmers · Gothenburg, Sweden

Aug 2026 – Jan 2027

Cryptography · Operating Systems · Logic in CS · Data Science & AI

CS50AI: Introduction to AI with Python — HarvardX (edX) *2025–26*

~100 hours · CS50 Certificate — all twelve projects, from search and logic to CNNs and NLP

Classes Préparatoires — 2-year scientific program · Pau, France *2023–2025*

French Baccalauréat — Maths & Computer Science (NSI), Expert Maths option *2023*

Graduated with Highest Honors (Mention Très Bien)

SELECTED PROJECTS

CYEDIN — Internship Management Platform — Backend & Auth, Laravel/React/PostgreSQL *2026*

- Built the backend and authentication of a web platform connecting students, companies and school staff across five user roles, in a 3-person team.
- Designed a single login with role-based access, and a forced password change on first login so administrators never know users' passwords. Deployed to Railway; code public on GitHub.

Facial Recognition with PCA / Eigenfaces — Technical Lead, Java/JavaFX *2025–26*

- Led architecture, code conventions, and Git/branch strategy for a 5-person team; owned integration across all modules.
- Implemented the projection and identification algorithms, including a three-criterion recognise/reject decision (Hotelling's T^2, reconstruction error, nearest-distance) calibrated empirically on a validation set.
- Cut computation on 40,000-dimension images by computing eigenfaces from the reduced covariance matrix (A^TA), and evaluated accuracy with confusion matrices.

Crowd Evacuation Simulator — 3-person team, Python *2025*

- Built a social-force agent model (NumPy) computing pairwise repulsion forces and obstacle avoidance, animated in real time with Tkinter.
- Implemented and compared Euler vs. Runge-Kutta integration against the analytical solution to quantify numerical error.

Samedis Neige — Web App for a Sports Association, PHP *2025*

- Wrote the CSV processing engine that generated attendance sheets and ski-level groups from association data files.
- Delivered a session-secured web app (upload, processing, dated history) for the 1,213-member Amicale Laïque de Billère; full lifecycle from needs analysis to demo.

EXPERIENCE

Private Tutor — Toulouse & Pau *2023–2024*

Tutored middle- and high-school students; built communication and client trust.

Facilities Maintenance Assistant — Cugnaux *2023*

Maintained municipal facilities to hygiene/safety standards; developed autonomy.

SKILLS

PROGRAMMING Python, Java, C, PHP, JavaScript/TypeScript, SQL

WEB Laravel, React, PostgreSQL, HTML/CSS

TOOLS Git, Linux

LANGUAGES French (native), English (fluent), German (school level), Norwegian (beginner)

INTERESTS Cybersecurity (PYR-CTF 2025 capture-the-flag), combat sports, theology